Lesson 1:
Overview of Human and Computer Interaction 1

What is HCI?

Human Computer Interaction is a field of study focused on interfaces between users and computers. HCI observes the way how humans associate with computers to produce efficient and useful interaction, examining and observing some factors based on how humans use technology and establish designs and programs to enable applicable systems.

User Interaction Design

Interaction design is about creating interventions in often complex situations using technology of many kinds including PC software, the web and physical devices.

User Interaction Design

Design involves:

- Achieving goals within constraints and trade-off
- Understanding the raw materials: computer and human
- Accepting limitations of human of design

The design process has several stages and is iterative and never complete.

Interaction starts with getting to know the users and their context:

- **Finding out who they are and what they are like …**
- **Talking to them, watching them**

User Interaction Design

- Scenarios are rich design stories, which can be used and reused throughout design:
  - They help us see what users will want to do
  - They give a step-by-step walkthrough of users interactions: including what they see, do and are thinking.

User Interaction Design

- Users need to find their way around a system; this involves:
  - helping users know where they are, where they have been and what they can do next.
  - creating overall structures that are easy to understand and fit the users needs.
  - designing comprehensible screens and control panels

User Interaction Design

- Complexity of design means we don’t get it right first time:
  - So, we need iteration and prototypes to try out and evaluate
  - But iteration can get trapped in local maxima, designs that have no simple improvements, but are not good
  - Theory and models can help give good start points

Usability

It is a measure of the effectiveness, efficiency and satisfaction with which specified user can achieve specified goals in a particular environment.

It asks the following:

- Is effective to use
- Is efficient to use
- Is safe to use
- Has good utility
- Is easy to learn
- Is easy to remember how to use

Usability

Issues in Design

- Who are the users?
- What do we mean by needs?
- How do generate alternative design?
- How do we choose between designs?

User Centered Development

- Data Collection
- Data Analysis
- User Modeling
- Design
- Prototyping
- Evaluation

- In User Modeling, a computational model for how people perform tasks and solve problems based on psychological principles.

- In design phase, all the analysis are used to design a system in interface design, it shows how this product present itself. And in interaction design, it tells how this product should work.

- Rapid prototypes are early and inexpensive ways to identify usability problems before committing lots of resources. In interaction design main concern here is with usability. So, using prototypes we can better understands the needs of user.

- In usability testing and evaluation phase, user perform a variety of task with a prototype (or other system) while observes record notes on what each user does and speaks.

- Typical tests are conducted with one user at a time or two users working together. Testing may include collecting data on the paths users take to do tasks, the errors they make, when and where they are confused or frustrated, how fast they do a task, whether they succeed in doing the task, and how satisfied they are with the experience.

- The goal of most usability testing is to uncover any problems that users may encounter so those problems can be fixed.

Interaction Styles

Interaction can be seen as dialogue between the computer and the user. Some applications have distinct styles of interaction.

We can identify some common styles

- Command Line Interface
- Menus
- Question/answer and query dialogue
- Form- fills and spreadsheet
- WIMP(Windows, Icon, Menus, Pointer)

Windows

Areas of the screen that behave as if they were independent terminals

- can contain text or graphics
- can be moved or resized
- can overlap and obscure each other, or can be laid out next to one another
- scrollbars allow the user to move the contents of the window up and down or from side to side
- title bars describe the name of the window

Icons

Small picture or image, used to represent some object in the interface, often a window. Windows can be closed to this small representation (iconized) allowing many windows to be accessible. Icons can be many and various – highly stylized or realistic representations.

Pointers

Important component, since WIMP style relies on pointing things such as icons and menu items.

- usually achieved with mouse
- joystick, trackball, cursor keys or keyboard shortcuts are also used.
- wide variety

END

Thank you and God bless!